Solution to Assignment 2, 2024

Q1:

Suppose that money is required to purchase both consumption and investment goods. The CIA constraint then becomes $c_t + x_t \leq \frac{m_{t-1}}{1+\pi_t} + \tau_t$, where x_t is the investment. Assume that the aggregate production function takes the form $y_t = e^{z_t} k_{t-1}^{\alpha} n_t^{1-\alpha}$, where n_t is the labor supply of each household. Show that the steady-state capital-labor ratio is affected by the rate of inflation. Does a rise in inflation raise or lower the steady-state capital-labor ratio? Explain.

Solution:

Consider the deterministic CIA model where the representative household solves the following lifetime utility maximization problem

$$\max_{c_t, n_t} \sum_{t=0}^{\infty} \beta^t u(c_t, 1 - n_t), \tag{2.1.1}$$

subject to the budget constraint

$$c_t + m_t + b_t + x_t \leq y_t + \frac{m_{t-1} + (1 + i_{t-1}) b_{t-1}}{1 + \pi_t} + \tau_t, \tag{2.1.2}$$

where the production function is given by

$$y_t = f(k_{t-1}, n_t) = e^{z_t} k_{t-1}^{\alpha} n_t^{1-\alpha}, \tag{2.1.3}$$

the CIA constraint

$$c_t + x_t \leq \frac{m_{t-1}}{1 + \pi_t} + \tau_t, \tag{2.1.4}$$

and the capital accumulation process

$$k_t = (1 - \delta) k_{t-1} + x_t. \tag{2.1.5}$$

The Bellmand equation is given by

$$V(k_{t-1}, m_{t-1}, b_{t-1}) = \max_{c_t, n_t, x_t, m_t, b_t, k_t} \{u(c_t, 1 - n_t) + \beta V(k_t, m_t, b_t)\}. \tag{2.1.6}$$

Let λ_t, μ_t and ρ_t be the Lagrange multipliers for (2.1.2), (2.1.4) and (2.1.5) respectively, then the first-order conditions with respect to consumption, labor, investment, money, bonds and capital are given by

$$u_c(c_t, 1 - n_t) = \lambda_t + \mu_t, \tag{2.1.7}$$

$$u_l(c_t, 1 - n_t) = \lambda_t f_n(k_{t-1}, n_t), \tag{2.1.8}$$

$$\rho_t = \lambda_t + \mu_t, \tag{2.1.9}$$

$$\beta V_m(k_t, m_t, b_t) = \lambda_t, \tag{2.1.10}$$

$$\beta V_b(k_t, m_t, b_t) = \lambda_t, \tag{2.1.11}$$

$$\beta V_k(k_t, m_t, b_t) = \rho_t. \tag{2.1.12}$$

The Envelop Theorem implies

$$V_k(k_{t-1}, m_{t-1}, b_{t-1}) = \lambda_t f_k(k_{t-1}, n_t) + \rho_t(1 - \delta), \tag{2.1.13}$$

$$V_m(k_{t-1}, m_{t-1}, b_{t-1}) = \frac{\lambda_t + \mu_t}{1 + \pi_t}, \tag{2.1.14}$$

$$V_b(k_{t-1}, m_{t-1}, b_{t-1}) = \frac{\lambda_t(1 + i_{t-1})}{1 + \pi_t}. \tag{2.1.15}$$

Therefore, FOCs (2.1.10) to (2.1.12) can be written as

$$\beta \frac{\lambda_{t+1} + \mu_{t+1}}{1 + \pi_{t+1}} = \lambda_t, \tag{2.1.16}$$

$$\beta \frac{\lambda_{t+1}(1 + i_t)}{1 + \pi_{t+1}} = \lambda_t, \tag{2.1.17}$$

$$\beta[\lambda_{t+1} f_k(k_t, n_{t+1}) + \rho_{t+1}(1 - \delta)] = \rho_t. \tag{2.1.18}$$

In the steady state, we have

$$u_c(c, 1-n) = \lambda + \mu, \tag{2.1.19}$$

$$u_l(c, 1-n) = \lambda(1-\alpha)k^{\alpha}n^{-\alpha}, \tag{2.1.20}$$

$$\rho = \lambda + \mu, \tag{2.1.21}$$

$$\beta\frac{\lambda+\mu}{1+\pi} = \lambda, \tag{2.1.22}$$

$$\beta\frac{\lambda(1+i)}{1+\pi} = \lambda, \tag{2.1.23}$$

$$\beta[\lambda\alpha k^{\alpha-1}n^{1-\alpha} + \rho(1-\delta)] = \rho. \tag{2.1.24}$$

By Eq. (2.1.21), (2.1.22) and (2.1.23), we obtain

$$\rho = \lambda + \mu = \frac{\lambda(1+\pi)}{\beta}. \tag{2.1.25}$$

Then substituting ρ into Eq. (2.1.24) yields

$$\beta\left[\lambda\alpha(k/n)^{\alpha-1} + \frac{\lambda(1+\pi)}{\beta}(1-\delta)\right] = \frac{\lambda(1+\pi)}{\beta}, \tag{2.1.26}$$

which implies

$$\frac{k}{n} = \left\{\frac{\beta^2\alpha}{[1-\beta(1-\delta)](1+\pi)}\right\}^{\frac{1}{1-\alpha}}, \tag{2.1.27}$$

Hence, the steady-state capital-labor ratio is decreasing in inflation. To understand this, note that the CIA constraint implies that investment goods have to be purchased by money carried from the last period. An increase in inflation raises the opportunity cost of money and thus the cost of investment. Hence, firms will adjust to a lower capital-labor ratio in production.

Q2:

Consider the following model:

$$\text{Preferences: } \mathbb{E}_t \sum_{i=0}^{\infty} \beta^i [\ln c_{t+i} + \theta \ln d_{t+i}],$$

$$\text{Budget constraint: } c_t + d_t + m_t + k_t = Ak_{t-1}^{\alpha} + \tau_t + \frac{m_{t-1}}{1+\pi_t},$$

$$\text{CIA constraint: } c_t \leq \frac{m_{t-1}}{1+\pi_t} + \tau_t,$$

where m_t denotes real money balances and π_t is the inflation rate from period $t-1$ to period t. The two consumption goods, c and d, represent cash (c) and credit (d) goods. The net transfer τ is viewed as a lump-sum payment (or tax) by the household.

1. Does this model exhibit superneutrality? Explain.

2. What is the rate of inflation that maximizes steady-state utility?

Solution:

The Bellman equation is given by

$$V(k_{t-1}, m_{t-1}) = \max_{c_t, d_t, m_t, k_t} \{\ln(c_t) + \theta \ln(d_t) + \beta V(k_t, m_t)\}. \tag{2.2.1}$$

Let λ_t and μ_t be the Lagrange multiplier for the budget constraint and CIA constraint respectively. Then the first-order conditions with respect to cash goods consumption, credit goods consumption, money, and capital are given by

$$\frac{1}{c_t} = \lambda_t + \mu_t, \tag{2.2.2}$$

$$\frac{\theta}{d_t} = \lambda_t, \tag{2.2.3}$$

$$\beta V_k(k_t, m_t) = \lambda_t, \tag{2.2.4}$$

$$\beta V_m(k_t, m_t) = \lambda_t. \tag{2.2.5}$$

The Envelop Theorem implies

$$V_k(k_{t-1}, m_{t-1}) = \lambda_t \alpha A k_{t-1}^{\alpha-1}, \tag{2.2.6}$$

$$V_m(k_{t-1}, m_{t-1}) = \frac{\lambda_t + \mu_t}{1 + \pi_t}. \tag{2.2.7}$$

Then the FOCs (2.2.4) and (2.2.5) can be written as

$$\beta \frac{\lambda_{t+1} + \mu_{t+1}}{1 + \pi_{t+1}} = \lambda_t, \tag{2.2.8}$$

$$\beta \lambda_{t+1} \alpha A k_t^{\alpha-1} = \lambda_t. \tag{2.2.9}$$

Hence, in the steady state, we have

$$\frac{1}{c} = \lambda + \mu, \tag{2.2.10}$$

$$\frac{\theta}{d} = \lambda, \tag{2.2.11}$$

$$\beta \frac{\lambda + \mu}{1 + \pi} = \lambda, \tag{2.2.12}$$

$$\beta \lambda \alpha A k^{\alpha-1} = \lambda. \tag{2.2.13}$$

Eq. (2.2.13) implies $k = (\beta \alpha A)^{1/(1-\alpha)}$, which is independent of inflation. And thus the goods market clearing condition $c + d = Ak^{\alpha} - k$ implies the total consumption is independent of inflation. However, Eq. (2.2.10) to (2.2.12) implies

$$\frac{d}{c} = \frac{\theta(1 + \pi)}{\beta}. \tag{2.2.14}$$

A higher inflation raises the ratio between credit goods and cash goods. Given the fact that total consumption does not change, this implies that individuals substitute away from cash goods toward credit goods. This is because the higher inflation gives rise to higher opportunity cost of holding money which is needed to purchase cash goods. Therefore, money is not superneutral in this model.

Since the total consumption is not affected by inflation, the utility can be affected

by inflation only through the allocation between cash goods and credit goods. Thus, to maximize the steady-state utility, the optimal inflation should give rise to an allocation such that the marginal utility of cash goods equals that of credit goods, i.e., $1/c = \theta/d$, or

$$\frac{d}{c} = \theta. \tag{2.2.15}$$

Therefore, the optimal inflation solves

$$\theta = \frac{\theta(1+\pi^*)}{\beta}, \tag{2.2.16}$$

which implies $\pi^* = \beta - 1 < 0$, which is actually a deflation.

Q3: Walsh, 4th Edition, Chap 3 Question 3.6.

6. Modify the basic model of section 3.3.1 by assuming utility depends on the consumption of two goods, C_t^m and C_t^c. Purchases of C_t^m are subject to a cash-in-advance constraint; purchases of C_t^c are not. The two goods are produced by the same technology: $C_t^m + C_t^c = Y_t = f(k_t)$.

   a. Write the household's decision problem.

   b. Write the first-order conditions for the household's optimal choices for C_t^m and C_t^c. How are these affected by the cash-in-advance constraint?

   c. Show that the nominal rate of interest acts as a tax on the consumption of C_t^m.

(1) The household's decision problem is

$$\max \quad \sum_{t=0}^{\infty} \beta^t u(c_t^m, c_t^c) \tag{21}$$

$$\text{s.t.} \quad c_t^m + c_t^c + k_t + b_t + m_t = f(k_{t-1}) + (1-\delta)k_{t-1} + \frac{b_{t-1}(1+i_{t-1})}{1+\pi_t} + \frac{m_{t-1}}{1+\pi_t} + \tau_t,$$

$$c_t^m \leq \frac{m_{t-1}}{1+\pi_t} + \tau_t,$$

(2) FOCs are

$$u_1(c_t^m, c_t^c) = \lambda_t + \mu_t, \tag{22}$$

$$u_2(c_t^m, c_t^c) = \lambda_t, \tag{23}$$

$$\lambda_t = \beta\lambda_{t+1}[f'(k_t) + 1 - \delta], \tag{24}$$

$$\lambda_t = \beta\lambda_{t+1}\left(\frac{1+i_t}{1+\pi_{t+1}}\right), \tag{25}$$

$$\lambda_t = \beta\left(\frac{\lambda_{t+1} + \mu_{t+1}}{1+\pi_{t+1}}\right). \tag{26}$$

From the above equations, we have

$$\frac{u_1(c_t^m, c_t^c)}{u_2(c_t^m, c_t^c)} = 1 + i_{t-1}. \tag{27}$$

When $i_{t-1} > 0$, CIA constraint is binding, i.e., $\mu_t > 0$. To reflect a higher marginal cost for C_m relative to C_c, i.e., $1 + i_{t-1}$ vs. 1, at the optimum, marginal benefit for C_m should be proportionally higher than that for C_c.

(3) Given that $\mu_{t+1} = \lambda_{t+1} i_t$, we have

$$u_1(c_t^m, c_t^c) = \lambda_t + \mu_t = \lambda_t(1 + i_t) \geq \lambda_t. \tag{28}$$

Thus, a positive nominal interest rate acts as a tax on consumption c^m; it raises the price of consumption c^m above its production cost. While consumption c^c doesn't imply such tax.

Q4:

$$1 = \lambda\beta \sum_{j=l,h} \pi_j \frac{1}{d + x_j}$$

$$\frac{1}{\alpha\beta} = \frac{\pi_l}{d + x_l} + \frac{\pi_h}{d + x_h}$$

$$\frac{d + \pi_l x_h + \pi_h x_l}{(d + x_l)(d + x_h)} = \frac{1}{\alpha\beta}$$

$$d^2 + (x_l + x_h - \beta l)d + (x_l + x_h - \beta\alpha[\pi_l x_h + \pi_h x_l]) = 0$$

$$d = \frac{1}{2}\left(\sqrt{(x_l + x_h - \beta l)^2 - 4(x_l + x_h - \beta\alpha[\pi_l x_h + \pi_h x_l])} - (x_l + x_h - \beta l)\right)$$

MSc Q5: Walsh, 4th Edition, Chap 3 Question 3.7.

7. Assume the model of section 3.3.1 is modified so that only a fraction ψ of consumption must be purchased using cash. In this case, the cash-in-advance constraint takes the form

$$\psi c_t \leq \frac{m_{t-1}}{1 + \pi_t} + \tau_t, 0 < \psi \leq 1.$$

a. Write the household's decision problem.

b. Write the household's first-order conditions. How are these affected by ψ?

c. If ψ were a choice variable of the household, would it ever choose $\psi > 0$?

Only a fraction ψ of consumption suffers CIA constraint.
The CIA constraint is

$$\psi c_t \leqslant \frac{m_{t-1}}{1 + \pi_t} + \tau_t, 0 < \psi \leqslant 1$$

(a). The problem is identical to the one in the textbook (3.21) except that ψ is part of CIA constraint.

(b). And the FOC (3.23) is slightly changed

$$u_c(c_t) - \lambda_t - \psi\mu_t = 0$$

In the textbook case, $\psi = 1$ and the cost of consumption includes both the marginal value of income λ_t and the cost of holding money to purchase the consumption goods μ_t. In this exercise, only a fraction ψ of consumption need to purchase in advance. The cost of holding case, being a part of consumption cost, is $\psi\mu_t$.

(c). If ψ is a choice variable, household will never choose $\psi > 0$.Since holding cash, as a purpose of consumption, is costly. The household should aviod any consumption need to be purchased using cash.

PhD Q5: Walsh, 4th Edition, Chap 3 Question 3.11.

Consider the following model. Preferences are given by

$$\mathbb{E}_t \sum_{i=0}^{\infty} \beta^i [\ln c_{t+i} + \ln d_{t+i}], \tag{2.3.1}$$

and the budget constraint is

$$c_t + d_t + m_t + k_t = Ak_{t-1}^a + \tau_t + \frac{m_{t-1}}{1+\pi_t} + (1-\delta)k_{t-1}, \tag{2.3.2}$$

where m_t denotes real money balances, and π_t is the inflation rate from period $t-1$ to period t. Utility depends on the consumption of two types of good; c must be purchased with cash, whereas d can be purchased using either cash or credit. The net transfer τ is viewed as a lump-sum payment (or tax) by the household. If a fraction θ of d is purchased using cash, then the household also faces a CIA constraint of the form

$$c_t + \theta d_t \leq \frac{m_{t-1}}{1+\pi_t} + \tau_t. \tag{2.3.3}$$

What is the relationship between the nominal rate of interest and whether the CIA constraint if binding? Explain. Will the household ever use cash to purchase d (i.e., will the optimal θ ever be greater than zero)?

Solution:

To bring out the nominal interest rate explicitly, we modify the budget constraint so that individuals can choose the amount of bond holding. This will not affect the results since in equilibrium each individual holds zero bonds by symmetry. Then the Bellman equation is given by

$$V(k_{t-1}, m_{t-1}, b_{t-1}) = \max_{c_t, d_t, m_t, b_t, k_t} \{\ln(c_t) + \ln(d_t) + \beta V(k_t, m_t, b_t)\}, \tag{2.3.4}$$

subject to the budget constraint

$$c_t + d_t + m_t + k_t + b_t = Ak_{t-1}^a + \tau_t + \frac{m_{t-1} + (1 + i_{t-1})b_{t-1}}{1 + \pi_t} + (1 - \delta)k_{t-1}, \tag{2.3.5}$$

and the CIA constraint

$$c_t + \theta d_t \leq \frac{m_{t-1}}{1 + \pi_t} + \tau_t, \tag{2.3.6}$$

where $0 \leq \theta \leq 1$.

Let λ_t and μ_t be the Lagrange multipliers for the budget constraint and CIA constraint respectively, and $\rho_1 \geq 0$, $\rho_2 \geq 0$ be the multiplier for the choice of θ. Then the first-order conditions with respect to cash goods, credit goods, money, capital, bonds

and θ are given by

$$\frac{1}{c_t} = \lambda_t + \mu_t, \tag{2.3.7}$$

$$\frac{1}{d_t} = \lambda_t + \theta\mu_t, \tag{2.3.8}$$

$$\beta V_m(k_t, m_t, b_t) = \lambda_t, \tag{2.3.9}$$

$$\beta V_k(k_t, m_t, b_t) = \lambda_t, \tag{2.3.10}$$

$$\beta V_b(k_t, m_t, b_t) = \lambda_t, \tag{2.3.11}$$

$$-\mu_t d_t + \rho_1 - \rho_2 = 0, \rho_1\theta = 0, \rho_2(1-\theta) = 0. \tag{2.3.12}$$

The Envelop Theorem implies

$$V_k(k_{t-1}, m_{t-1}, b_{t-1}) = \lambda_t[aAk_{t-1}^{a-1} + (1-\delta)], \tag{2.3.13}$$

$$V_m(k_{t-1}, m_{t-1}, b_{t-1}) = \frac{\lambda_t + \mu_t}{1+\pi_t}, \tag{2.3.14}$$

$$V_b(k_{t-1}, m_{t-1}, b_{t-1}) = \frac{\lambda_t(1+i_{t-1})}{1+\pi_t}. \tag{2.3.15}$$

Hence, FOCs (5.1.8) to (2.3.11) can be written as

$$\beta\frac{\lambda_{t+1} + \mu_{t+1}}{1+\pi_{t+1}} = \lambda_t, \tag{2.3.16}$$

$$\beta\lambda_{t+1}[aAk_t^{a-1} + (1-\delta)] = \lambda_t, \tag{2.3.17}$$

$$\beta\frac{\lambda_{t+1}(1+i_t)}{1+\pi_{t+1}} = \lambda_t. \tag{2.3.18}$$

In the steady state, we have

$$\frac{1}{c} = \lambda + \mu, \tag{2.3.19}$$

$$\frac{1}{d} = \lambda + \theta\mu, \tag{2.3.20}$$

$$\beta\frac{\lambda + \mu}{1 + \pi} = \lambda, \tag{2.3.21}$$

$$\beta\lambda[aAk^{a-1} + (1 - \delta)] = \lambda, \tag{2.3.22}$$

$$\beta\frac{\lambda(1 + i)}{1 + \pi} = \lambda, \tag{2.3.23}$$

$$\rho_1 - \rho_2 = \mu d. \tag{2.3.24}$$

By Eq. (2.3.21) and (2.3.23), we obtain

$$i = \frac{\mu}{\lambda}. \tag{2.3.25}$$

Hence, as long as $i > 0$, we have $\mu > 0$ and the CIA constraint is binding. Note that the nominal interest rate measures the opportunity cost of holding money. Therefore, as long as holding money is costly, the consumer will hold as less money as possible and hence the CIA constraint must be binding.

By Eq. (2.3.24), we have $\rho_1 > 0$ since one of ρ_1 and ρ_2 must be zero and $\rho_1 - \rho_2 = \mu d > 0$. Therefore the optimal $\theta = 0$, i.e., the consumer will never use cash to purchase d. This is understandable because holding cash is costly.

MSc & PhD Q6: Walsh, 4th edition, Chap 5 Question 5.6.

Following the notation in the textbook, let λ_1, λ_2, and λ_3 denote the Lagrangian multipliers associated with cash-in-advance constraint, budget constraint, and wages-in-advance constraint, respectively. The FOCs are

$$
\begin{aligned}
E_{t-1}[\lambda_{1t}] &= E_{t-1}[R_t\lambda_{2t}], && (40)\\
u'(c_t) &= \lambda_{1t} + \lambda_{2t}, && (41)\\
v'(n_t^s) &= \omega_t\lambda_{2t}, && (42)\\
\lambda_{2t}Y'(n_t^d) &= \omega_t(\lambda_{2t} + \lambda_{3t}), && (43)\\
\lambda_{3t} &= R_t\lambda_{2t}, && (44)\\
\lambda_{2t} &= \beta E_t\left[\frac{\lambda_{1t} + \lambda_{2t}}{1 + \pi_{t+1}}\right]. && (45)
\end{aligned}
$$

If ignoring uncertainty, we have

$$\frac{v'(n_t^s)}{u'(c_t)} = \frac{\omega_t\lambda_{2t}}{\lambda_{1t} + \lambda_{2t}} = \frac{\omega_t}{(\lambda_{1t}/\lambda_{2t}) + 1} = \frac{\omega_t}{1 + R_t}. \tag{46}$$

Also, from FOC, we have

$$Y'(n_t^d) = \omega_t(1 + \lambda_{3t}/\lambda_{2t}) = \omega_t(1 + R_t). \tag{47}$$

Therefore, the marginal rate of substitution between leisure and consumption equals the marginal product of labor divided by $(1 + R_t)^2$

$$\frac{v'(n_t^s)}{u'(c_t)} = \frac{Y'(n_t^d)}{(1 + R_t)^2}. \tag{48}$$

The intuition is that: at the optimum, the marginal cost of increasing one unit of labor, $v'(n_t^s)$, equals to the marginal benefit from the following amount of consumption: marginal product of labor $Y'(n_t^d)$ first discounted by $(1 + R_t)$ due to firm paying interests from wages-in-advance constraint and then discounted by another $(1 + R_t)$ due to household facing opportunity cost from cash-in-advance constraint.